

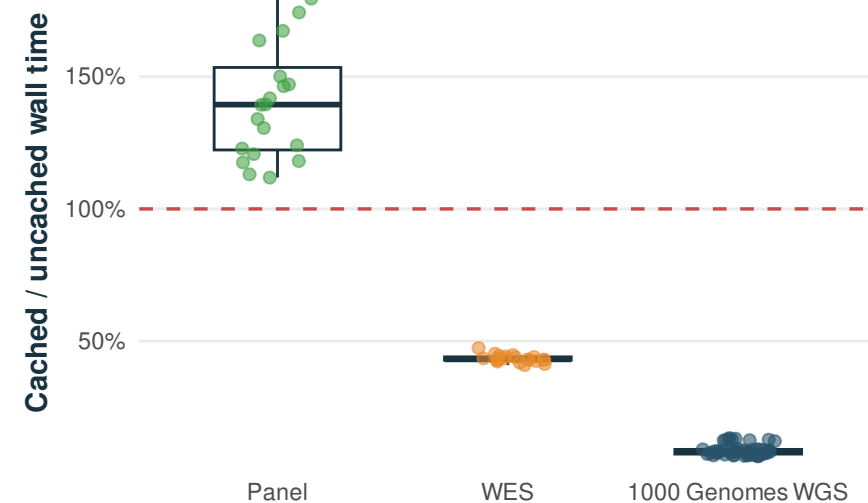
VCFcache impact across assay sizes

Bundled Zenodo gnomAD AF $\geq 1\%$ cache · VEP 115.2 --everything · one paired run per sample

A

Time remaining after caching

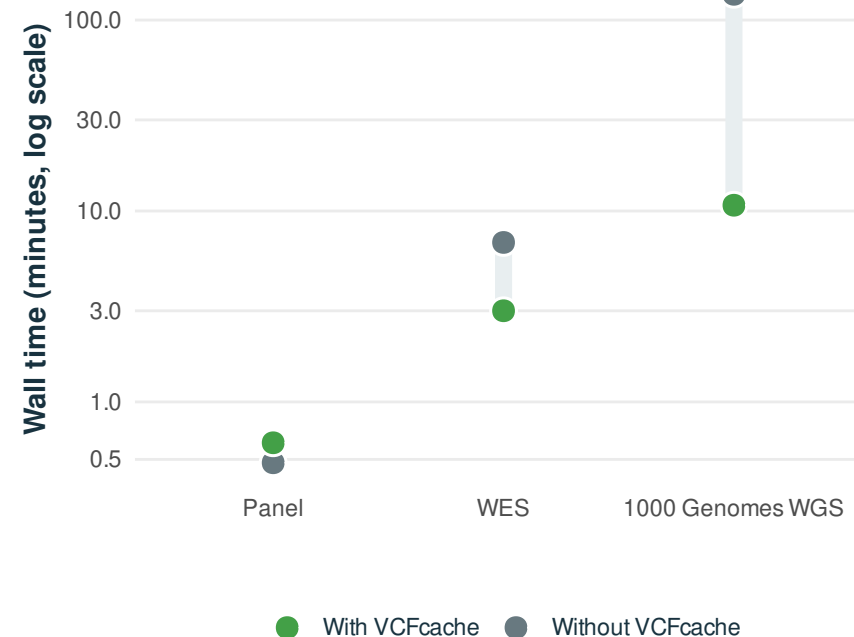
Each point is one biological sample; line = no runtime benefit



B

Median wait per sample

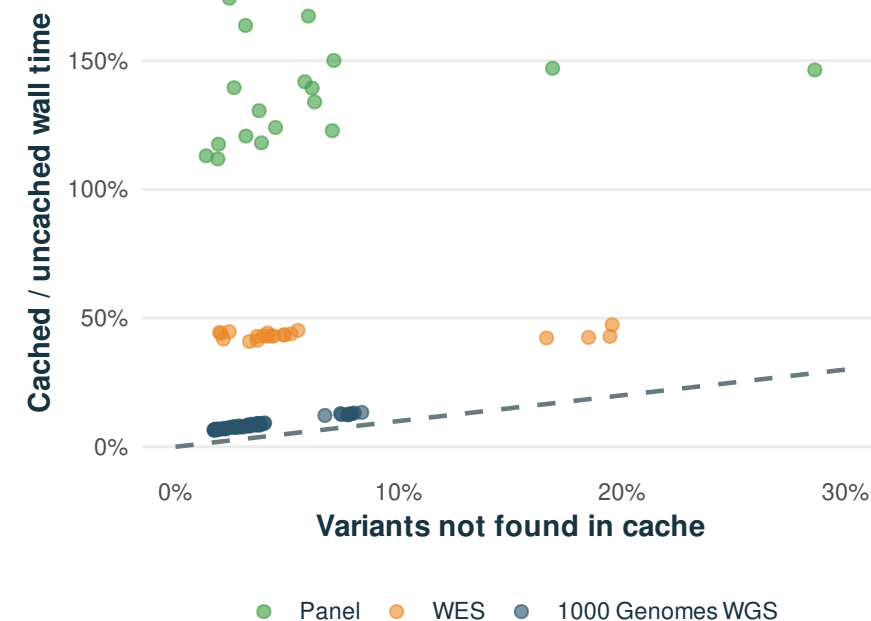
The panel pipeline is too short for lookup overhead to pay back



C

Misses drive annotation work

Dashed line: zero-overhead runtime rule



D

✓ **89/89 paired outputs pass**

Unexpected annotation mismatches: 0 | Known upstream VEP HGNC_ID differences reported separately: 265